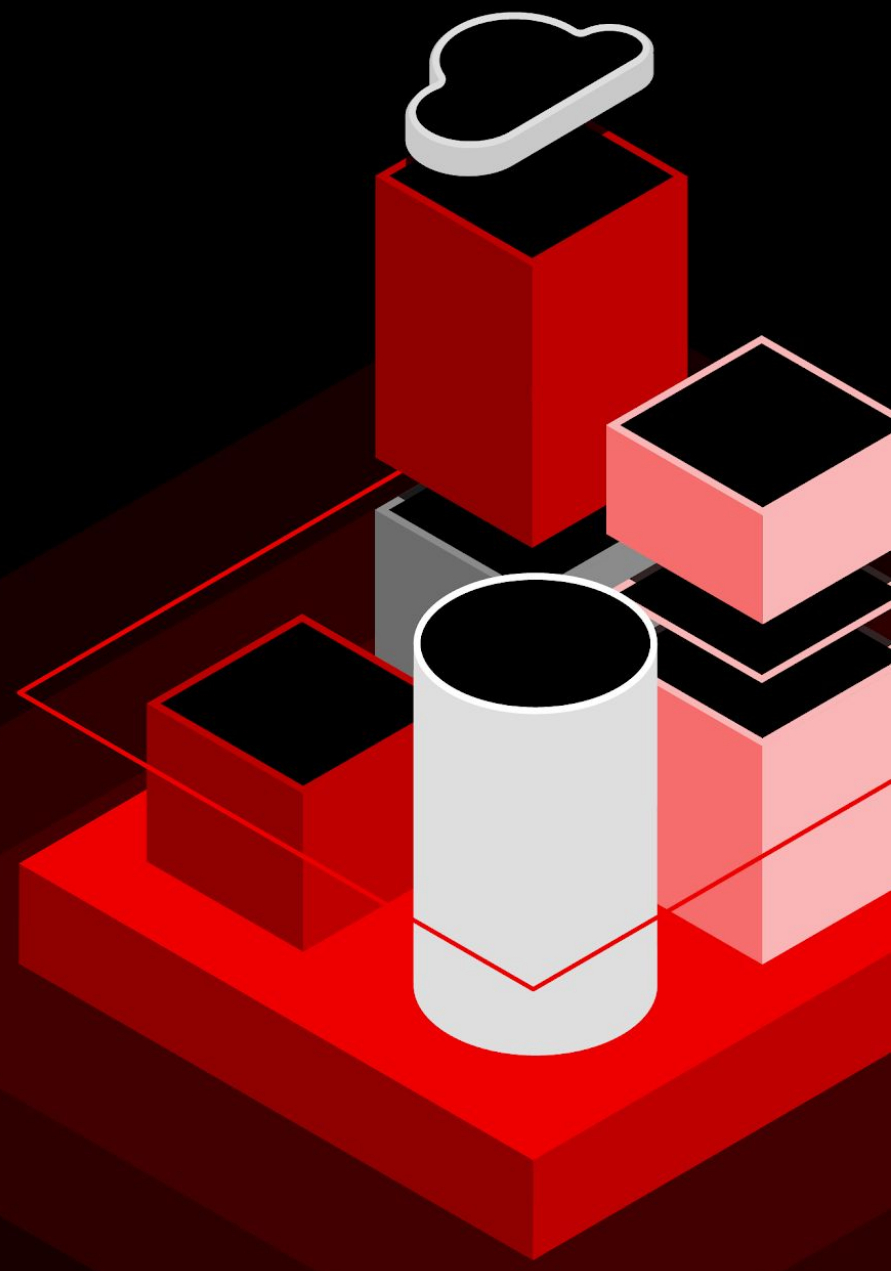
The Convergence of HPC, K8s and (increasingly) AI

A lot happening, let's break it down

Kush Gupta
Senior Solution Architect

Wes Jackson
Senior Specialist Solution Architect, AI



How did we get here?

Traditional batch vs. Cloud-native

Then

- Monolithic MPI Applications
- Static job scripts
- Exclusive allocations
- Dedicated file systems

Now

- The LLMs
- Loosely coupled AI workflows
- Streaming and distributed data
- Model checkpoints, inference APIs
- Iterative tuning & evaluation
- Artifacts to consistently store and iterate on experiment data

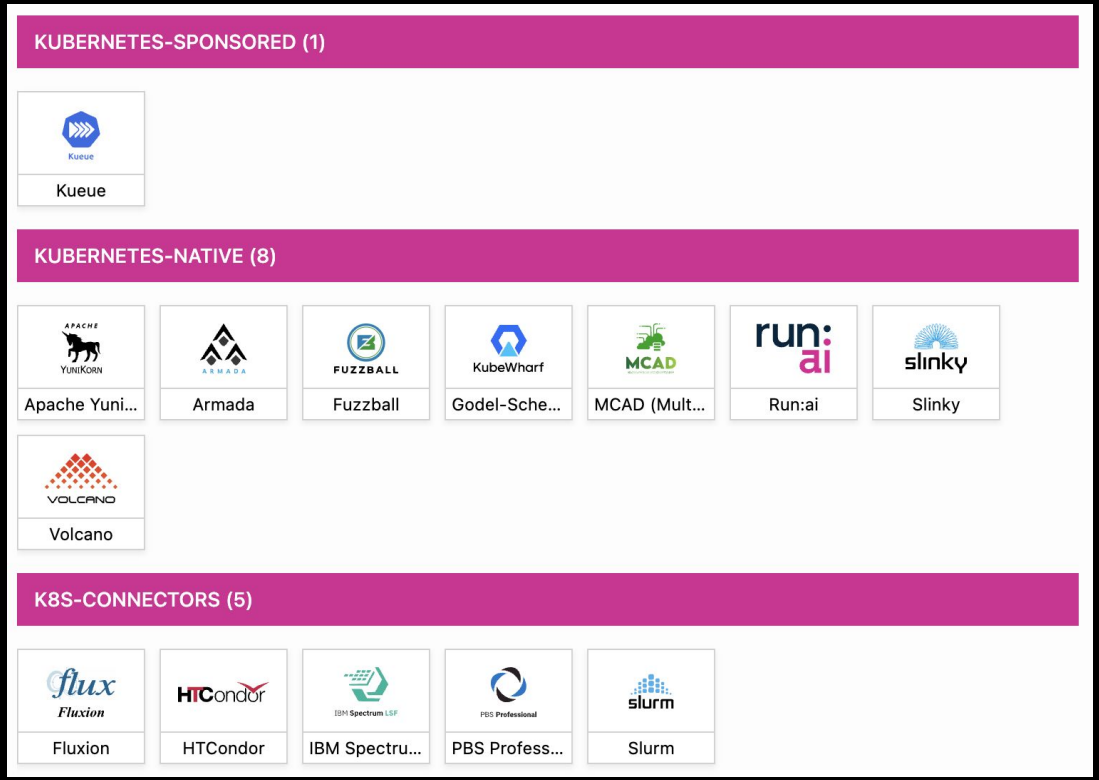
Source:
Badia, Rosa M., Berti-Equille, Laure, Ferreira Da Silva, Rafael, & Leser, Ulf (2024). Integrating HPC, AI, and Workflows for Scientific Data Analysis: Report from Dagstuhl Seminar 23352. https://doi.org/10.2172/2341398

Where are we exactly?

A journey in progress

Primary Approaches

- **Add support for Kubernetes into existing HPC schedulers**
- Build new cloud-native batch schedulers on Kubernetes



Source:
https://tag-runtime.cncf.io/wgs/bsi/landscape/

Why focus on existing HPC schedulers?

Build on what is already there

- Existing scientists/users & admins will (rightly) **drop-kick us** if we say "go 100% to K8s, you'll be fine!"
  - Nothing will work as it exactly did before, no one will be happy
- Switching platforms *alone* will never solve the root goal (desire best platform for the right *thing*)
  - The tech, culture and processes takes time to grow & evolve
- Must retain functional parity of both HPC software and hardware
  - Broader ecosystem of parallel file systems, specialized HPC hardware and more to ensure compatibility
  - kube-scheduler lacks fundamental awareness of future utilization (backfill), and lacks full topology awareness compared to existing HPC schedulers

Starting somewhere **familiar** provides a strong foundation for improvement

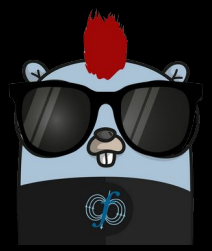
Not all inclusive of course!

Two relevant projects

From some of the best

Flux Framework

- Creator: LLNL
- Flux can run as a parallel job under most launchers that support MPI, including under itself or Slurm
- **Flux Operator**
  - K8s Operator to create and control a Flux MiniCluster to launch jobs to



Slinky

- Creator: SchedMD
- Integrating Slurm capabilities with Kubernetes
- **Slurm Operator**
  - Deploying Slurm onto Kubernetes in pods
- Slurm Bridge
  - Integration between existing Slurm clusters and Kubernetes for cross-scheduling

Source:
https://flux-framework.readthedocs.io/en/latest/
https://flux-framework.org/flux-operator/
https://www.schedmd.com/slinky/why-slinky/
https://slurm.schedmd.com/SLUG23/flux_in_slurm.pdf

Where do we want to go?

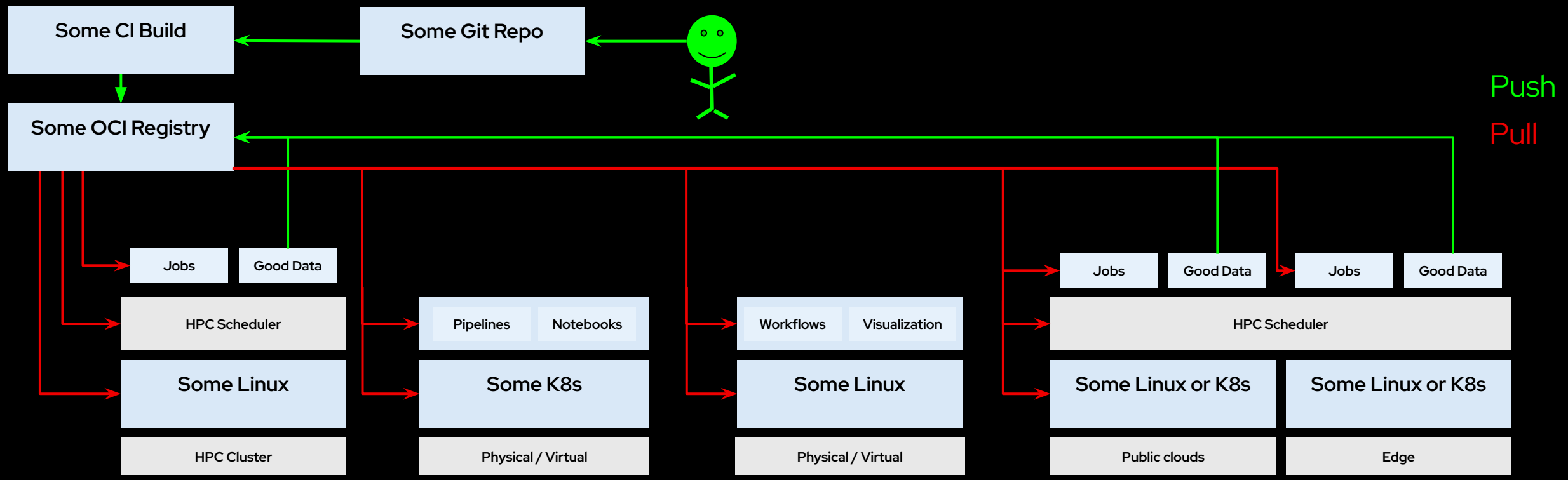
More science, everywhere!

Lots of things can be containers and managed through OCI registries now

- Applications/processes as containers for a while
- Getting an entire, immutable Operating System as containers
  - Through bootc
  - Can include custom libraries & kernel modifications for HPC environments
    - Diskless support/implementation is an open discussion
- We can also manage AI models, files, helm, and other data from experiments as containers
  - Through oras, and more recently podman artifact!
- And now we have HPC schedulers as containers (oversimplification)

What does this get us?

Services spread out among clusters and platforms everywhere, right?



Reference:
Jason Kincl's Slurm and Batch Computing on OpenShift Presentation

What's the catch?

Fragmentation and Compute Compatibility

- An out-of-the-box, OCI-compliant runtime is generic *by design*
  - No default awareness of HPC hardware or software (Accelerators, interconnects, MPI libraries)
  - Implementation of different runtimes (Podman, Apptainer, Kubernetes) and execution across them for same applications are inconsistent
    - Wasn't the whole point of containers that consistency?!

- Awareness of the system topology and matching the right flags/holes to the right jobs are required for portable scale

https://docs.nersc.gov/development/containers/podman-hpc/overview/

A path!

How has this been approached before?

Account for & provide that awareness!

podman-hpc

https://github.com/NERSC/podman-hpc

Module Name	Function
--mpi	Uses optimized Cray MPI
--cuda-mpi	Uses CUDA-aware optimized Cray MPI
--gpu	Enable NVIDIA GPU
--nccl	Enables NCCL OFI plugin for CUDA 12 containers
--cvmfs	Enable the CVMFS filesystem
--openmpi-pmi2	Helper module for PMI2 support in OpenMPI
--openmpi-pmix	Helper module for PMIx support in OpenMPI

Ramalama

https://github.com/containers/ramalama

REPOSITORY NAME

- ramalama / ramalama
- ramalama / ramalama-rag
- ramalama / cuda
- ramalama / rocm
- ramalama / ramalama-llama-server
- ramalama / smollm
- ramalama / llama-stack
- ramalama / rocm-rag
- ramalama / asahi
- ramalama / cuda-rag
- ramalama / ramalama-whisper-server
- ramalama / intel-gpu
- ramalama / cuda-llama-server
- ramalama / openvino
- ramalama / musa
- ramalama / cann
- ramalama / intel-gpu-rag
- ramalama / ramalama-cli
- ramalama / stable-diffusion

https://docs.nersc.gov/development/containers/podman-hpc/overview/#podman-at-nersc
https://github.com/containers/ramalama

Rama-what?

Ramalama!

Containerization enables **modularity**, easily swapping out for new blocks

Model Registry Choice

Hugging Face

OPEN CONTAINER INITIATIVE

?

Inference Runtime Choice

LLaMA C++

wHISper C++

vLLM

?

MLX

No Merriam-Webster here

What defines an artifact?

Blobs

```
❯ vim super-cool.file
❯ podman artifact add quay.io/kugupta/cool:today super-cool.file …
88e9a985f746bd74b84392031400d029a3c1f78cc27bf047730b3f534df5cb9e
❯ podman artifact inspect quay.io/kugupta/cool:today
{
    "Manifest": {
        "schemaVersion": 2,
        "mediaType": "application/vnd.oci.image.manifest.v1+json", # identifies OCI image manifest
        "config": {
            "mediaType": "application/vnd.oci.empty.v1+json", #NOT a container image
            "digest": "sha256:44136fa355b3678a1146ad16f7e8649e94fb4fc21fe77e8310c060f61caaff8a",
            "size": 2,
            "data": "e30="
        },
        "layers": [
            {
                "mediaType": "text/plain; charset=utf-8",
                "digest": "sha256:dd0e685ba3372ad7507fc2d53cdad81e9c2a06ce113884a0230a9ba9a4cd2e81",
                "size": 20,
                "annotations": {
                    "org.opencontainers.image.title": "super-cool.file" # original filename tracked
                }
            }
        ]
    },
    "Name": "quay.io/kugupta/cool:today",
    "Digest": "sha256:88e9a985f746bd74b84392031400d029a3c1f78cc27bf047730b3f534df5cb9e"
}
❯ podman artifact push quay.io/kugupta/cool:today
```

https://docs.podman.io/en/stable/markdown/podman-artifact.1.html

What can be an artifact?

Whatever you want!

- The CNCF ModelPack Specification standardizes packaging, distributing, and running AI models in cloud-native environments
  - Provides annotations for:
    - Model architecture, format (onnx, tensorflow, pytorch etc.) size, precision and quantization
    - Input and output types, knowledge cutoff, reasoning, tool use, languages
    - Weights, documentation, datasets
    - UID, GID, file permissions, last time modified

- Can link the model to the **exact pipeline** that created/finetuned it, params and all
- Allows you to inherit all benefits of OCI:
  - Granular, team-based auth
  - Portability
  - Immutability
  - Security scanning & SBOMs
  - Signing and verification

https://github.com/modelpack/model-spec/blob/main/docs/spec.md

Where they at!?

in-progress

kubernetes / enhancements

Code | Issues 269 | Pull requests 86 | Actions | Projects | Security | Insights

VolumeSource: OCI Artifact and/or Image #4639

Open

sallyom opened on May 16, 2024 · edited by saschagrunert — Edits — Contributor

Enhancement Description

- One-line enhancement description (can be used as a release note): VolumeSource: OCI Artifact or OCI Image
- Kubernetes Enhancement Proposal: https://github.com/kubernetes/enhancements/tree/master/keps/sig-node/4639-oci-volume-source
- Discussion Link:
- Primary contact (assignee): sgrunert@redhat.com, somalley@redhat.com
- Responsible SIGs: sig-node, sig-storage
- Enhancement target (which target equals to which milestone):
  - Alpha release target (x.y): 1.31
  - Beta release target (x.y): 1.33-1.34
  - Stable release target (x.y):

- [x] Alpha
  - [x] KEP (`k/enhancements`) update PR(s):
    - KEP-4639: adding OCI VolumeSource #4642
  - [x] Code (`k/k`) update PR(s):
    - [KEP-4639] Add OCI VolumeSource CRI API kubernetes#125659
      - [KEP-4639] Update CRI API and workflow #4751
    - [KEP-4639] Add `ImageVolumeSource` API kubernetes#125660
    - [KEP-4639] Add `ImageVolumeSource` implementation kubernetes#125663
    - **Nice to have:**
      - [KEP-4639] Mention that `fsGroupChangePolicy` has no effect kubernetes#126281
      - Fix runtime panic in imagevolume `CanSupport` method kubernetes#126323
      - [KEP-4639] Add `ImageVolumeSource` node e2e tests kubernetes#126220
  - [x] Out of tree update PR(s):
    - Add OCI Volume Source support to `crictl [create|run] --with-pull` kubernetes-sigs/cri-tools#1464
    - Add OCI/Image Volume Source support cri-o/cri-o#8317
    - Make ImageVolume garbage collection work cri-o/cri-o#8408
    - CRI-O: add `pull-kubernetes-node-crio-cgrpv2-imagevolume-e2e` test job test-infra#33071
    - Fix imagevolume test focus test-infra#33088
  - [x] Docs (`k/website`) update PR(s):
    - Document OCI volume sources / KEP-4639 website#46946
    - Blog post about OCI volume sources / KEP-4639 website#46960
- [x] Beta (disabled by default)
  - [x] KEP (`k/enhancements`) update PR(s):
    - KEP-4639: update to most recent implementation state and graduate to beta #4897
    - 4639: bump to 1.34 and document noexec drop #5417
    - [KEP-4639] Image Volume: remove noexec option #5354
    - 4639: bump to 1.34 and document noexec drop #5417 **retargeting beta in v1.34, dropped noexec requirement**
  - [x] Code (`k/k`) update PR(s):
    - [KEP-4639] Graduate image volume sources to beta kubernetes#130135
    - Enable `ImageVolume` tests for `pull-kubernetes-node-e2e-containerd-alpha-features` test-infra#34398
    - *Optional:* [KEP-4639] Image volume subpath e2e kubernetes#130681
  - [x] Docs (`k/website`) update(s):
    - [KEP-4639] Graduate image volume source to beta website#49936
    - [KEP-4639] Feature blog 1.33: Image volumes graduate to beta website#49984
  - [x] Out of tree update PR(s):
    - Add image volume subpath support cri-o/cri-o#9050
    - Update cri-api to latest commit kubernetes-sigs/cri-tools#1793
    - Bump CRI-O cgroups v2 jobs to `bdcf72` test-infra#34547
    - Fix CNI setup for CRI-O test-infra#34549
- [x] Beta (enabled by default)
  - [x] KEP (`k/enhancements`) update PR(s): [KEP-4639] Graduate image volumes to GA #5450
  - [x] Code (`k/k`) update PR(s):
    - [KEP-4639] Graduate image volumes to GA kubernetes#134502
    - KEP 4639: Move ImageVolume to on by default beta kubernetes#135195
    - [KEP-4639] Remove dedicated image volume presubmit test-infra#35668
  - [ ] Docs (`k/website`) update(s):
    - [KEP-4639] Graduate image volume enabled by default (beta) website#52733
- Follow-ups:
  - Read-write volumes as outlined in: KEP-4639: Allow writeable, container isolated volume content #5288

containers / ramalama

Code | Issues 49 | Pull requests 12 | Discussions | Actions | Projects | Wiki | Security | Insights

[WIP] Add support for converting to OCI artifacts #2046

Open — rhatdan wants to merge 1 commit into `containers:main` from `rhatdan:artifact`

Conversation 45 | Commits 1 | Checks 34 | Files changed 11

rhatdan commented last week • edited by sourcery-ai bot — Member

Just showing this off, I am still working on PRs to get into podman 5.7 release.

containers/podman#27329
containers/podman#27328
containers/podman#27325
containers/podman#27324
containers/podman#27319
containers/podman#27253

Summary by Sourcery

Add comprehensive support for storing and managing AI models as OCI artifacts, extending the transport, CLI, and configuration layers, and covering the new functionality with extensive documentation and end-to-end system tests.

New Features:

- Add support for OCI 'artifact' type for model conversion, push, pull, list, inspect, and removal operations
- Introduce CLI options and configuration for specifying 'artifact' conversion type alongside 'car' and 'raw'

Enhancements:

- Extend OCI transport to handle artifact-specific commands (_add_artifact, _rm_artifact, inspect, mount_cmd, is_artifact)
- Integrate artifact listing into oci_tools and combine with existing image and manifest listings
- Update base transport for correct mounting and inspect behavior based on artifact flag
- Allow default conversion type to be set via configuration and environment variable with correct precedence

Documentation:

- Update command-line help and man pages to include 'artifact' type and describe its behavior
- Add configuration documentation for the new 'convert_type' setting

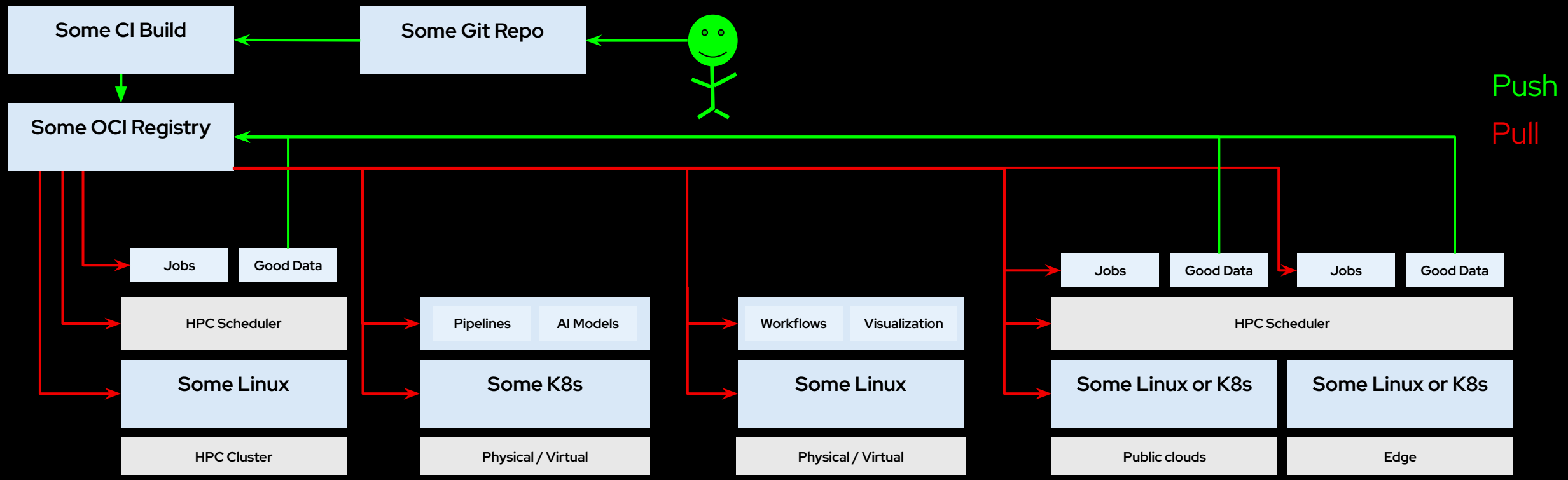
Tests:

- Add comprehensive system tests for artifact workflows including conversion, push, pull, listing, removal, error handling, configuration precedence, and size reporting

Source:
https://github.com/kubernetes/enhancements/issues/4639
https://github.com/containers/ramalama/pull/2046

Actually in sight!

Services spread out among clusters and platforms everywhere!



Reference:
Jason Kincl's Slurm and Batch Computing on OpenShift Presentation

On November 30, 2022, OpenAI released ChatGPT

As a free public "research preview," which quickly went viral on social media.
The launch made advanced AI accessible to the public,
leading to massive user adoption (over a million users in less than a week)

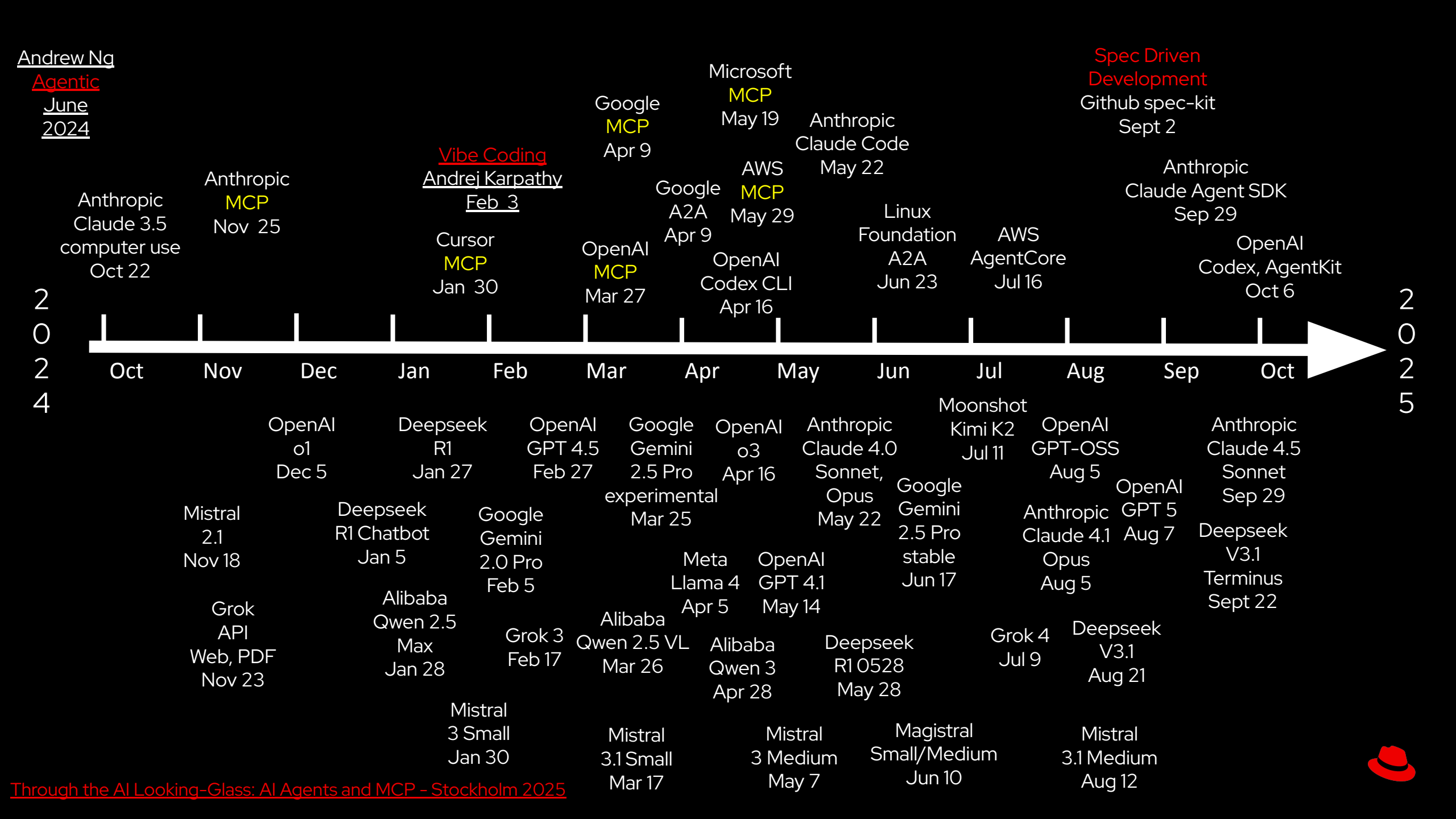
Gartner Says Worldwide AI Spending Will Total $1.5 Trillion in 2025

STAMFORD, Conn., September 17, 2025

Continuous Demand for AI Is Expected to Fuel Investments in IT Infrastructure

Worldwide spending on AI is forecast to total nearly $1.5 trillion in 2025 according to Gartner, Inc. a business and technology insights company.

https://www.gartner.com/en/newsroom/press-releases/2025-09-17-gartner-says-worldwide-ai-spending-will-total-1-point-5-trillion-in-2025


Andrew Ng
Agentic
June
2024
Anthropic
Claude 3.5
computer use
Oct 22
Anthropic
MCP
Nov 25
Vibe Coding
Andrej Karpathy
Feb 3
Cursor
MCP
Jan 30
Google
MCP
Apr 9
OpenAI
MCP
Mar 27
Google
A2A
Apr 9
Microsoft
MCP
May 19
AWS
MCP
May 29
OpenAI
Codex CLI
Apr 16
Anthropic
Claude Code
May 22
Linux
Foundation
A2A
Jun 23
AWS
AgentCore
Jul 16
Spec Driven
Development
Github spec-kit
Sept 2
Anthropic
Claude Agent SDK
Sep 29
OpenAI
Codex, AgentKit
Oct 6
2024
Oct
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Aug
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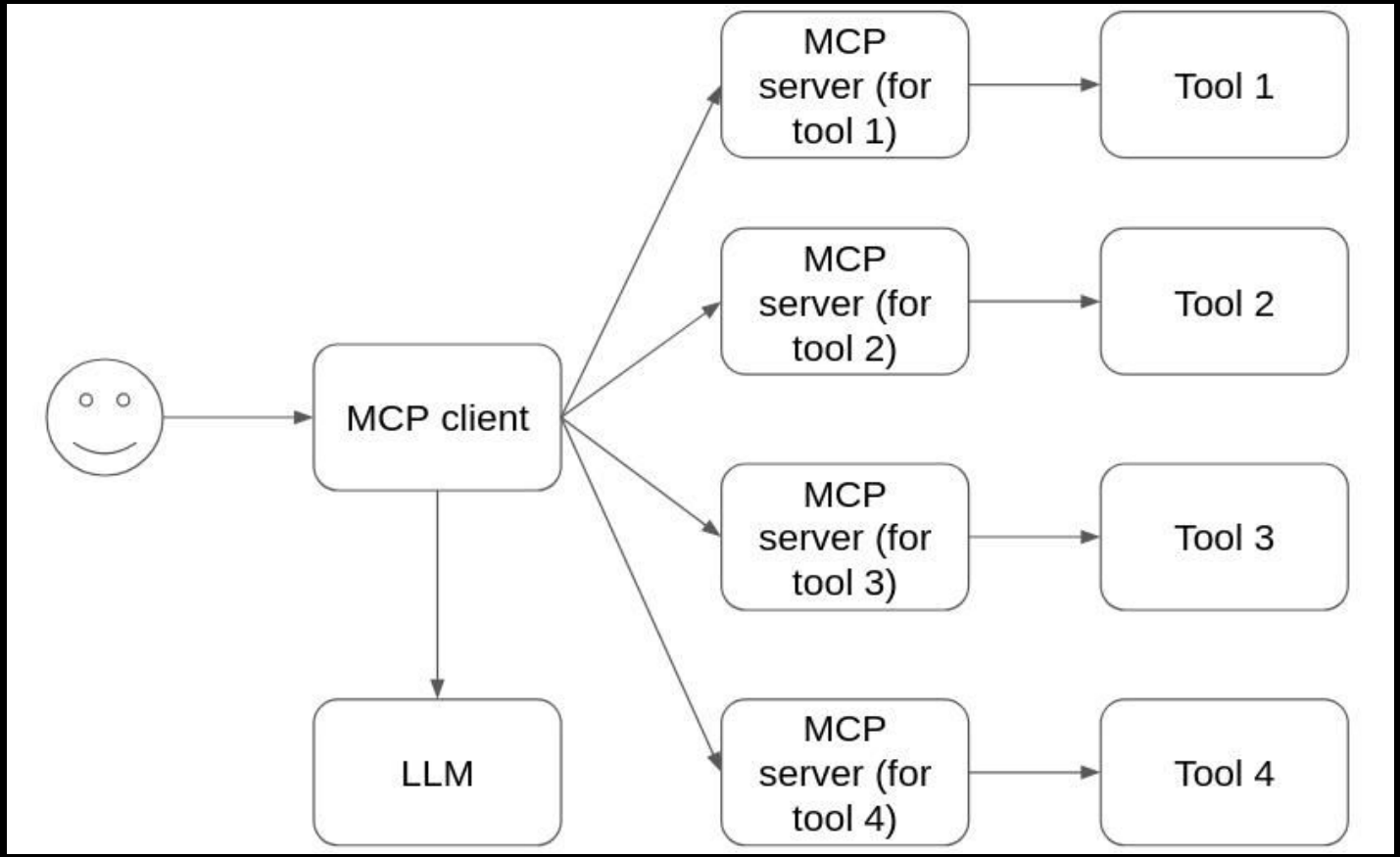
Oct
2025
OpenAI
o1
Dec 5
Deepseek
R1
Jan 27
OpenAI
GPT 4.5
Feb 27
Google
Gemini
2.5 Pro
experimental
Mar 25
OpenAI
o3
Apr 16
Anthropic
Claude 4.0
Sonnet,
Opus
May 22
Moonshot
Kimi K2
Jul 11
OpenAI
GPT-OSS
Aug 5
Anthropic
Claude 4.5
Sonnet
Sep 29
Mistral
2.1
Nov 18
Deepseek
R1 Chatbot
Jan 5
Google
Gemini
2.0 Pro
Feb 5
Google
Gemini
2.5 Pro
stable
Jun 17
Anthropic
Claude 4.1
Opus
Aug 5
OpenAI
GPT 5
Aug 7
Deepseek
V3.1
Terminus
Sept 22
Grok
API
Web, PDF
Nov 23
Alibaba
Qwen 2.5
Max
Jan 28
Grok 3
Feb 17
Alibaba
Qwen 2.5 VL
Mar 26
Meta
Llama 4
Apr 5
OpenAI
GPT 4.1
May 14
Alibaba
Qwen 3
Apr 28
Deepseek
R1 0528
May 28
Grok 4
Jul 9
Deepseek
V3.1
Aug 21
Mistral
3 Small
Jan 30
Mistral
3.1 Small
Mar 17
Mistral
3 Medium
May 7
Magistral
Small/Medium
Jun 10
Mistral
3.1 Medium
Aug 12
Through the AI Looking-Glass: AI Agents and MCP - Stockholm 2025

Model Context Protocol

A brief overview



https://www.redhat.com/en/blog/model-context-protocol-mcp-understanding-security-risks-and-controls

Is it secure?

The S in MCP stands for security

- Must enforce AuthN & AuthZ
  - OAuth 2.1
    - Enables an application to obtain limited access to a protected resource, either on **behalf of a resource owner** by orchestrating… **between the resource owner and an authorization service**, or by allowing the application to obtain access on its own behalf
    - Also need to ensure granular scope of access (what a user can do may not be what we want an MCP to do)
- Local servers could execute host OS commands, but remote MCP servers aren't perfect either
  - Supply chain risks, prompt injection, tool injection, sampling and more

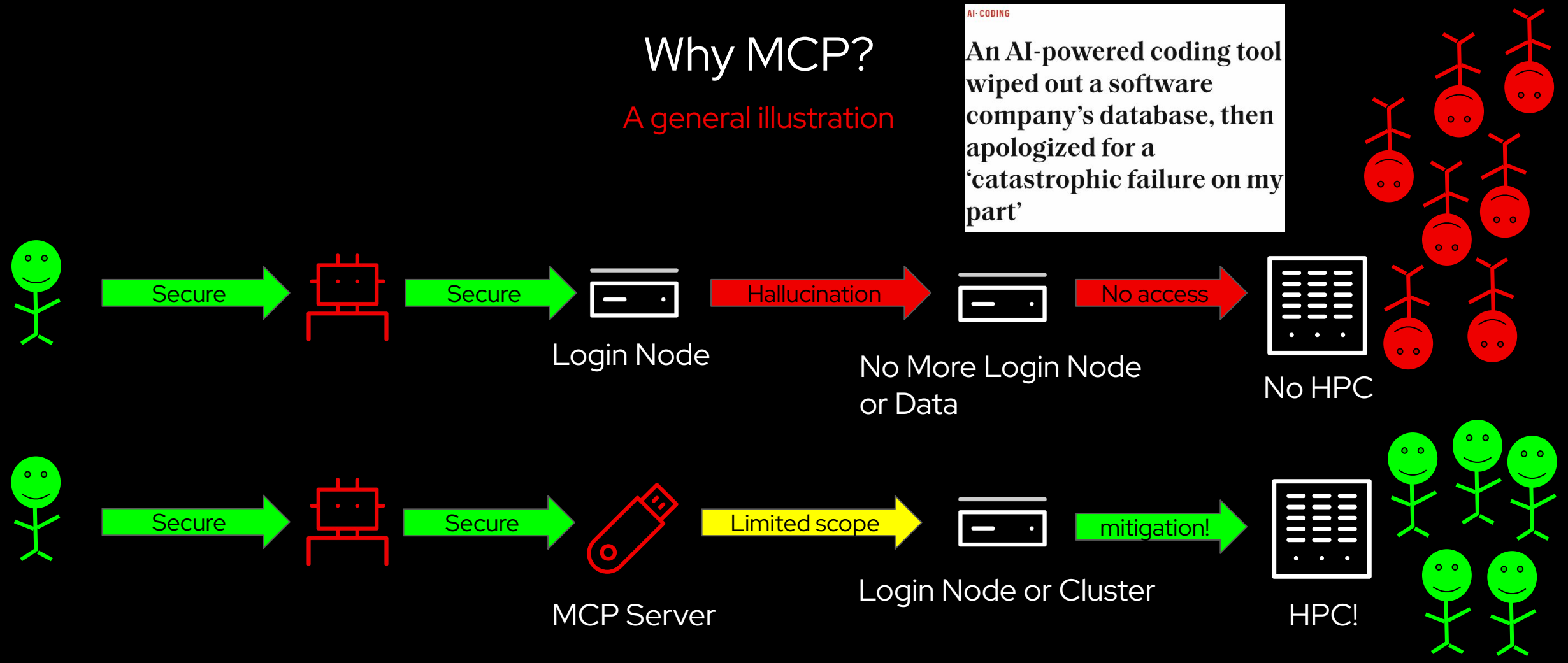
https://datatracker.ietf.org/doc/html/draft-ietf-oauth-v2-1-13
https://www.redhat.com/en/blog/model-context-protocol-mcp-understanding-security-risks-and-controls

Why MCP?

A general illustration

AI· CODING

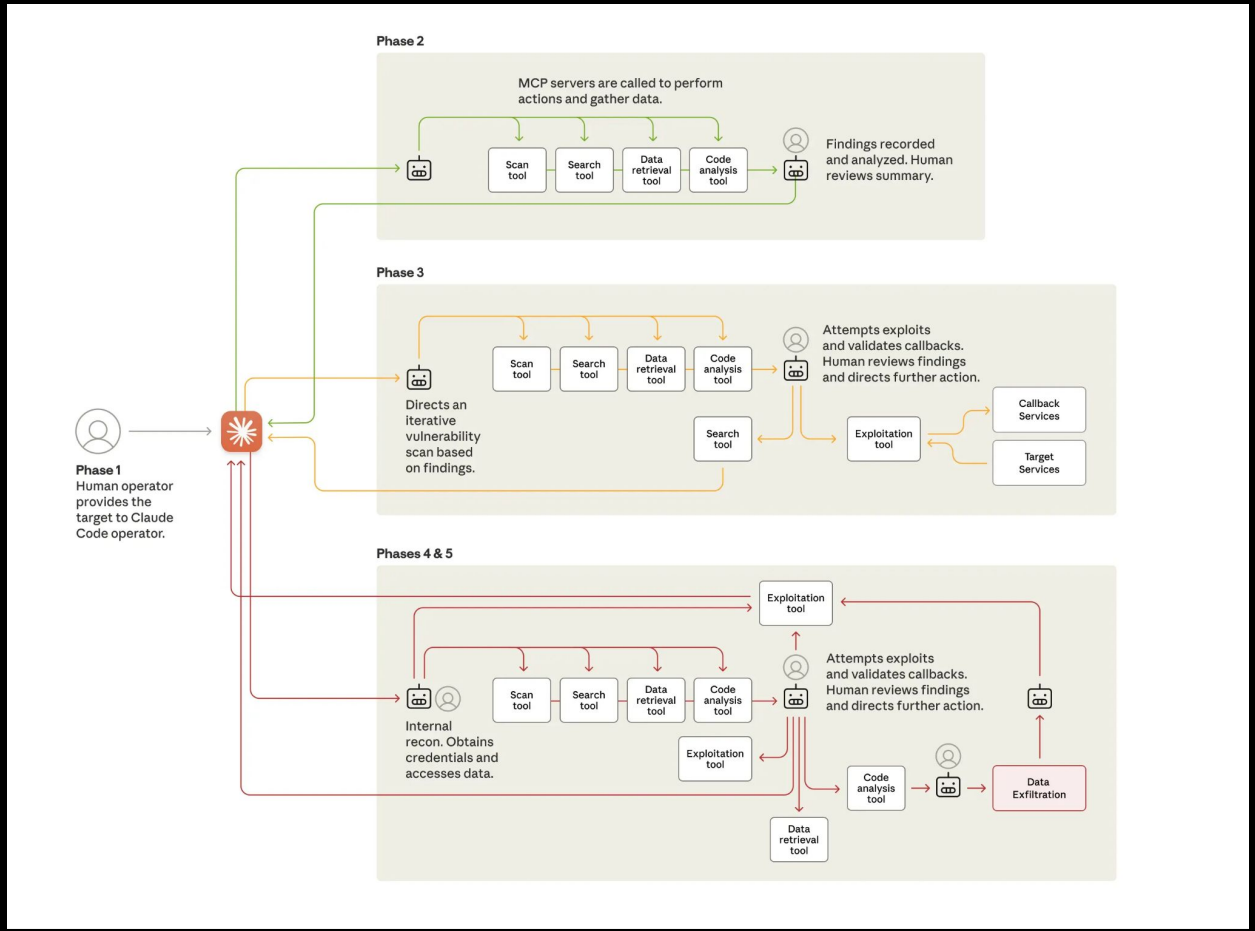
An AI-powered coding tool wiped out a software company's database, then apologized for a 'catastrophic failure on my part'



https://fortune.com/2025/07/23/ai-coding-tool-replit-wiped-database-called-it-a-catastrophic-failure/

Example use case?

Cyber espionage campaign!



https://www.anthropic.com/news/disrupting-AI-espionage

What could a HPC MCP server be?

Customized for your user environments

Generic/Template as a starting point

- Integrate with existing auth(s)
  - Limit scope for what it can execute on the behalf of a user
- Scheduler execution and info
- RBAC for what the server itself can do
- Can integrate with other tools, underlying platform, services outside of batch
- **Logging**

Customized for your community

- See what's being run!
  - In this specific AI scope
  - See what works well, and what does not
    - Create a baseline
- Customize/Finetune models towards closing gaps
- Insights can extend to components outside MCP (overall trends, patterns)
- Help new users better learn unfamiliar tooling

https://github.com/redhat-data-and-ai/template-mcp-server
https://github.com/rdwj/mcp-server-template

Gen AI Production Considerations & Best Practices

Security

- OAuth/SSO authentication
- Configure SSL/TLS certificates
- Use secure cookie signing
- Implement rate limiting
- Add input validation

Scalability

- Use managed DB
- Horizontal pod autoscaling
- Load balancing
- Caching strategies

Monitoring

- Set up agent tracing
- Logging aggregation
- Add health check monitoring
- Set up alerting

MCP Server

- Keep tools focused and single-purpose
- Include comprehensive error handling
- Add detailed logging
- Write tests for each tool
- Document tool parameters clearly

Agent

- Use streaming for better UX
- Implement conversation persistence
- Add tracing for debugging and logging
- Configure appropriate timeouts
- Handle tool failures gracefully

UI

- Implement proper loading states
- Show streaming responses
- Handle auth errors
- Optimize bundle size

Reference:
Red Hat Data and AI Team
https://www.ibm.com/downloads/documents/us-en/1443d5dd174f42e6

References

Links!

- Bootstrap Kind, an MCP server, Slurm & Flux Operator: https://github.com/kush-gupt/sc25
  - Reference MCP template:
    - https://github.com/rdwj/mcp-server-template

- red.ht/SC2025
  - Demo Schedule at Red Hat Booth **#2216**
    - Slurm and Slinky Operator on OpenShift demo - Tuesday, November 18 - 1:30 pm
    - Podman and Sarus Suite: cloud-native productivity meets native HPC speed - Wednesday, November 19 - 2:30 pm

Thank you

Red Hat is the world's leading provider of enterprise open source software solutions. Award-winning support, training, and consulting services make Red Hat a trusted adviser to the Fortune 500.

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